

Lab 8: Files

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May 13, 2026

Space Station Management

In this lab, continue the *Space Station Management* project and add support for reading and saving data in text and CSV files.

Basic problem

Prepare a CSV file named *crew_members.csv* containing crew member data. Each row should store:

- name,
- age,
- role,
- specialization,
- mission hours.

Add a static method *loadCrewMembersCSV* to the *SpaceStation* class (or another helper class). The method should read data from a given CSV file and return an *ArrayList<CrewMember>*.

If the file does not exist or contains incorrect data, the method should throw *IllegalArgumentException*.

In the main function:

- read data from *crew_members.csv*,
- create crew member objects,
- print all loaded crew profiles,
- calculate and display:
 - average age of the crew,
 - average number of mission hours,
 - number of crew members in each role.

Save the summary to a text file named *crew_report.txt*.

Space Station supplies

Prepare another CSV file named *supplies.csv*. Each row should store:

- supply name,
- quantity,
- cost of one unit.

Add a static method *loadSuppliesCSV* to the *SpaceStation* class (or another helper class). The method should read data from the CSV file and return an *ArrayList<Supply>*.

If the file does not exist or contains incorrect data, the method should throw *IllegalArgumentException*.

In the main function:

- read data from *supplies.csv*,
- create supply objects,
- calculate the total cost for each supply,
- calculate the total cost of all station supplies.

Save the results to a text file named *supplies_report.txt*.

Additional task

Create a method that saves all crew members assigned to one selected module to a separate CSV file.

For example, if the module name is *Science Lab*, save the result to *science_lab_crew.csv*.

The exported file should contain at least:

- crew member id,
- name,
- role,
- specialization,
- mission hours.

Hints

- Use *File*, *Scanner*, *FileWriter*, or *PrintWriter*.
- Split CSV rows with *split(",")*.
- Reuse your existing classes from previous Space Station labs.
- Keep the code clear and test both correct and incorrect input files.